



BLUESTOCK FINTECH PVT. LTD.

CAPSTONE INTERNSHIP PROJECT REPORT

Mutual Fund Analytics Platform: Data-Driven Performance Analysis & Interactive Business Intelligence Dashboard

Submitted By

Pramod Vinayak Patil

Organization:

Bluestock Fintech Pvt. Ltd.

Project Domain: Data Analytics | Mutual Fund Performance Analysis | Business Intelligence

Date: June 2026

Declaration

I, Pramod Vinayak Patil, student of B.E. (Electronics and Communication Engineering) at R.C. Patel Institute of Technology, Shirpur, hereby solemnly declare that the project report entitled:

"Mutual Fund Analytics Platform: Data-Driven Performance Analysis & Interactive Business Intelligence Dashboard"

submitted in partial fulfillment of the Bluestock Fintech Capstone Internship Programme is a record of original work carried out by me under the guidance and supervision of Bluestock Fintech Pvt. Ltd.

I further declare that:

- The work presented in this report has not been submitted to any other institution or university for the award of any degree, diploma, or certificate.
- All data, statistics, and figures presented in this report are sourced from publicly available and duly acknowledged sources, including AMFI India, mfapi.in, NSE India, and BSE India.
- The analysis, computations, and code contained herein are entirely my own work, developed independently using Python, SQL, and Power BI.
- All external references and sources of information have been duly cited in the References section.

Name: Pramod Vinayak Patil

Date: June 2026

Place: Shirpur, Maharashtra

Acknowledgement

The successful completion of this capstone project would not have been possible without the guidance, support, and encouragement of several individuals and institutions, to whom I extend my sincere gratitude.

First and foremost, I am deeply grateful to Bluestock Fintech Pvt. Ltd. for providing me with this exceptional internship opportunity. The structured project framework, technical resources, and industry-aligned problem statements offered by Bluestock Fintech have been invaluable in shaping my understanding of the financial technology domain.

I extend my heartfelt thanks to the project mentors and supervisors at Bluestock Fintech Pvt. Ltd. for their continuous guidance, constructive feedback, and encouragement throughout the project lifecycle. Their expert knowledge of the mutual fund industry and data analytics domain greatly enriched the depth and quality of this work.

My sincere appreciation goes to the open-data community, including AMFI India for maintaining a transparent and publicly accessible repository of mutual fund NAV data, and to the developers of mfapi.in for providing a clean, free API for historical NAV retrieval. The NSE and BSE India data portals similarly deserve acknowledgment for facilitating benchmark data access.

I also thank the open-source communities behind Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, SQLAlchemy, and the broader data science ecosystem without which this project would not have been technically feasible.

Finally, I express my deepest gratitude to my family and friends for their unwavering moral support, patience, and motivation throughout the duration of this internship and project.

— *Pramod Vinayak Patil*
June 2026

Executive Summary

This report presents a comprehensive account of the Mutual Fund Analytics Platform, a full-stack data analytics and business intelligence solution developed as part of the Bluestock Fintech Capstone Internship in June 2026. The project addresses a critical gap in the retail investment advisory landscape: the absence of an integrated, data-driven platform capable of delivering actionable mutual fund performance insights to investors, financial advisors, and wealth managers.

The Indian mutual fund industry has grown to an unprecedented scale, with Assets Under Management (AUM) crossing ₹81 lakh crore as of December 2025, and monthly SIP inflows recording an all-time high of ₹31,002 crore. Despite this rapid expansion, the majority of retail investors continue to rely on fragmented, manually curated data sources and simplified star-rating systems that fail to capture the multi-dimensional nature of fund performance. This project was conceived to directly address this structural inefficiency.

The platform ingests publicly available NAV data from AMFI India and mfapi.in, processes it through a robust Python ETL pipeline, and computes a comprehensive set of performance metrics — including Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR) — across 40 mutual fund schemes spanning 4.5 years of historical data (January 2022 to May 2026). The processed data is stored in a normalised SQLite database built on a star schema architecture, enabling efficient multi-dimensional querying.

A four-page interactive Power BI dashboard was developed to surface these insights in a visually compelling and analytically rigorous format, covering industry overview metrics, individual fund performance analytics, investor transaction patterns, and SIP and market trend analysis. The dashboard incorporates interactive slicers across all pages, enabling stakeholders to filter data by fund house, category, date range, and geographic region.

Key findings from the analysis include the identification of top-performing large-cap funds by risk-adjusted returns, significant performance dispersion within the same fund category, a strong positive correlation between Sharpe Ratio and three-year CAGR among equity funds, and notable regional disparities in SIP penetration across Indian states.

Dimension	Detail
Fund Schemes Analysed	40 schemes across Large Cap, Mid Cap, Small Cap, Hybrid, and Debt categories
Historical Coverage	January 2022 – May 2026 (4.5 years 87,000+ data rows)
Data Sources	AMFI India, mfapi.in, NSE India, BSE India
Technology Stack	Python (Pandas, NumPy, SciPy), SQLite, Power BI, Jupyter Notebooks
Metrics Computed	CAGR, Daily Returns, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Max Drawdown, VaR (95%), CVaR
Database Tables	8 tables in star schema (2 dimension, 3 fact tables)
Dashboard Pages	4 pages: Industry Overview, Fund Performance, Investor Analytics, SIP & Market Trends
Project Duration	7 working days (50–55 estimated hours)

Table ES-1: Project Summary at a Glance

The platform demonstrates measurable business value across multiple stakeholder segments. For retail investors, it democratises access to institutional-grade analytical tools. For financial advisors and wealth managers, it streamlines the fund comparison process, reducing research time by an estimated 60–70%. For investment firms, the modular ETL architecture provides a scalable foundation for enterprise-grade analytics infrastructure.

1. Introduction

1.1 Overview of Mutual Funds

A mutual fund is a professionally managed, collective investment vehicle that pools capital from a large number of investors and deploys it across a diversified portfolio of securities, including equities, bonds, money market instruments, and other asset classes. In India, mutual funds are regulated by the Securities and Exchange Board of India (SEBI) and governed operationally by the Association of Mutual Funds in India (AMFI). The primary value proposition of mutual funds lies in their capacity to offer retail investors access to professionally managed, diversified portfolios at relatively low cost — a feature that has historically been the exclusive domain of high-net-worth individuals and institutional investors.

Mutual funds are broadly classified into three categories: equity funds, which invest predominantly in stocks; debt funds, which invest in fixed income instruments; and hybrid funds, which maintain exposure to both equity and debt in varying proportions. Within each category, further sub-classifications exist based on market capitalisation (large-cap, mid-cap, small-cap), investment strategy (growth, value, index), and thematic orientation (sectoral, international). The Net Asset Value (NAV) of a mutual fund unit is computed daily and reflects the per-unit market value of the fund's underlying portfolio after deducting expenses.

1.2 Importance of Fund Performance Analysis

Fund performance analysis is the systematic process of evaluating a mutual fund's historical and projected return characteristics relative to risk, benchmark indices, and peer group funds. In an environment where the Indian mutual fund industry offers over 1,500 distinct schemes to retail investors, the ability to rigorously and objectively assess performance becomes not merely useful, but essential. Unsophisticated comparison based solely on headline returns can be deeply misleading — a fund delivering 18% annual returns while exhibiting extreme volatility and frequent drawdown events may, on a risk-adjusted basis, significantly underperform a fund generating 14% returns with consistent, low-volatility growth.

Professional fund performance analysis employs a suite of quantitative metrics, including CAGR for measuring compounded growth, the Sharpe Ratio for risk-adjusted return efficiency, the Sortino Ratio for downside-risk penalisation, Alpha and Beta for benchmark-relative performance, and Maximum Drawdown for worst-case scenario quantification. These metrics, when presented in an integrated analytical framework, provide investors with the comprehensive intelligence necessary to make genuinely informed investment decisions.

1.3 Industry Background

The Indian mutual fund industry has undergone a remarkable transformation over the past decade, evolving from a niche financial product for the sophisticated urban investor to a mainstream retail savings instrument with pan-India reach. The industry's AUM crossed the historic milestone of ₹50 lakh crore in December 2023 and has since continued its upward trajectory, reaching ₹81 lakh crore by December 2025.

The SIP (Systematic Investment Plan) mechanism has been particularly transformative. Monthly SIP inflows, which stood at approximately ₹8,000 crore in 2018, rose to ₹31,002 crore in December 2025 — an all-time high — demonstrating the deepening of financial literacy and disciplined investment culture among retail Indians. Total folios (investor accounts) reached 26.12 crore, with a significant share attributable to first-time investors from tier-2 and tier-3 cities. This democratisation of investment, while positive, also underscores the critical need for accessible, high-quality fund analysis tools that can assist the growing investor base in navigating the complex and often opaque mutual fund selection landscape.

2. Business Problem Statement

Despite the significant growth of the Indian mutual fund industry and the proliferation of investment platforms, retail investors continue to face substantial analytical challenges when making fund selection and portfolio allocation decisions. This project was motivated by five distinct, interrelated business problems observed in the contemporary investment advisory ecosystem.

2.1 Complexity of Large-Scale Fund Comparison

The Indian mutual fund marketplace currently offers over 1,500 registered schemes from more than 40 Asset Management Companies (AMCs). While regulatory bodies such as SEBI have standardised fund categorisation, the sheer volume of options creates a decision paralysis problem for retail investors and even for less-technologically-equipped independent financial advisors. Evaluating even a subset of 40–50 funds across multiple performance dimensions manually — using spreadsheets or fragmented online portals — is an exercise in extreme inefficiency. Investors lack a unified analytical layer capable of processing and presenting all relevant comparative data in a single, organised interface.

2.2 Difficulty in Interpreting Raw NAV Data

NAV (Net Asset Value) data, while publicly available through AMFI India, exists primarily in raw, time-series format — a stream of daily prices without context, trend analysis, or comparative benchmarking. A non-expert investor viewing historical NAV data for a fund cannot intuitively discern whether the fund has performed well relative to its category peers, whether its growth is consistent or volatile, or whether any observed gains are commensurate with the level of risk assumed. The transformation of raw NAV data into meaningful insights requires technical processing that is beyond the capabilities of the typical retail investor.

2.3 Absence of Integrated Risk-Return Evaluation

The most common error in retail mutual fund evaluation is the exclusive focus on absolute return figures — a fund that gained 25% in a year is superficially more attractive than one that gained 15%, regardless of the risk assumed to achieve that return. Academic finance and professional investment management have long established that the correct unit of performance evaluation is the risk-adjusted return — the return per unit of risk taken. Without access to metrics such as the Sharpe Ratio, Sortino Ratio, Standard Deviation, Beta, and Maximum Drawdown, an investor cannot perform a genuine apples-to-apples comparison between funds of similar mandate but different risk profiles.

2.4 Lack of a Single Decision-Making Dashboard

Even investors who are aware of multi-dimensional performance metrics typically encounter them in silos — return data on one platform, risk metrics on another, portfolio composition on a third, and peer comparison on a fourth. The cognitive overhead of synthesising disparate data from multiple sources significantly degrades decision quality and increases the time cost of investment research. A single, integrated business intelligence dashboard that presents all relevant performance

dimensions in one view — with interactive filtering capabilities — represents a critical gap in the current investment tools landscape.

2.5 Time-Intensive Nature of Fund Performance Research

For financial advisors, wealth managers, and institutional investors managing client portfolios across multiple fund categories, the manual process of fund performance research and comparison is prohibitively time-intensive. Research that might require 4–6 hours of manual data gathering and spreadsheet analysis can, with an appropriately designed analytical platform, be compressed into minutes. This efficiency gain translates directly into improved client service, more frequent portfolio reviews, and ultimately better investment outcomes for end investors.

3. Business Objectives

The Bluestock Fintech Mutual Fund Analytics Platform was designed and developed with the following specific, measurable, and industry-aligned business objectives:

- Develop a robust, automated ETL (Extract, Transform, Load) pipeline capable of ingesting raw NAV data from AMFI India and mfapi.in, performing comprehensive data cleaning and validation, and loading the processed data into a structured analytical database — reducing manual data preparation effort to near zero.
- Compute a complete suite of professional-grade mutual fund performance metrics — including CAGR, Annualised Return, Volatility, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR (95%), and CVaR — for 40 fund schemes spanning 4.5 years of historical data, with full methodological transparency and formula documentation.
- Design and implement a normalised star-schema SQLite database with 8 tables, supporting efficient multi-dimensional querying across fund, date, performance, and investor transaction dimensions.
- Develop a comprehensive, four-page interactive Power BI dashboard that presents industry-level KPIs, individual fund performance analytics, investor behaviour patterns, and SIP and market trend analysis, with cross-page interactive filtering via fund house, category, date range, and geographic slicers.
- Implement a rule-based fund recommender system capable of suggesting the top three funds by Sharpe Ratio for a given investor risk appetite (Conservative, Moderate, Aggressive).
- Produce a quantitative fund scorecard that ranks funds on a composite 0–100 scale incorporating three-year return rank, Sharpe Ratio rank, alpha rank, expense ratio (inverse), and maximum drawdown (inverse), enabling direct fund-to-fund comparability.
- Establish a replicable, well-documented analytical framework — comprising Jupyter notebooks, Python scripts, SQL queries, and a Power BI dashboard — that can be extended and scaled to accommodate additional fund schemes, longer historical periods, and more advanced analytical modules.

4. Scope of Project

4.1 Inclusions

The scope of this project encompasses the complete data lifecycle from raw NAV data ingestion to interactive dashboard visualisation. Specifically, the following activities and deliverables are within scope:

- Data ingestion from AMFI India (monthly NAV reports, AUM data, SIP flow data) and mfapi.in (historical daily NAV via JSON API) for 40 mutual fund schemes across large-cap, mid-cap, small-cap, hybrid, and debt categories.
- Data cleaning, validation, and preprocessing using Python and Pandas, including handling of missing values via forward-fill, normalisation of date formats, detection and removal of outliers, and merging of data from multiple source files.
- Computation of nine financial performance metrics per fund scheme: CAGR, Annualised Return, Daily Return, Annualised Volatility, Sharpe Ratio, Sortino Ratio, Alpha, Beta (relative to Nifty 100), and Maximum Drawdown.
- Advanced analytics including Rolling 90-Day Sharpe Ratio, Historical VaR (95%) and CVaR, Investor Cohort Analysis, SIP Continuation Analysis, and Sector Concentration (HHI).
- SQLite database design and implementation using a star schema with dimension tables (dim_fund, dim_date) and fact tables (fact_nav, fact_transactions, fact_performance).
- Ten analytical SQL queries covering top fund rankings, category-level performance aggregation, risk-return profiling, and investor behaviour analysis.
- Power BI dashboard development with four report pages, interactive slicers, KPI cards, scatter plots, bar charts, line charts, maps, and a sortable fund scorecard table.
- A rule-based fund recommender script providing investor-profile-aware fund recommendations.

4.2 Exclusions

The following activities and capabilities are explicitly outside the scope of the current project iteration:

- Real-time or live-streaming NAV updates during market hours (the platform uses end-of-day NAV data).
- Portfolio optimisation using Markowitz Efficient Frontier methodology (identified as a future enhancement).
- Machine learning-based predictive NAV forecasting or fund rating prediction.
- Direct integration with brokerage or trading platforms for live order execution.
- Coverage of international mutual funds, ETFs, Index Funds, and Fund of Funds beyond the 40 schemes analysed.
- Compliance or regulatory reporting features.

5. Literature Review

The academic and professional literature on mutual fund performance evaluation, data analytics in finance, and business intelligence dashboard development provides a rich theoretical foundation for this project. This chapter surveys the key works and frameworks that have informed the methodological design of the Bluestock Fintech Mutual Fund Analytics Platform.

5.1 Mutual Fund Performance Measurement

The foundation of modern mutual fund performance analysis rests on the risk-return framework developed by Markowitz (1952) and subsequently operationalised by Sharpe (1966), whose Reward-to-Variability Ratio — now universally known as the Sharpe Ratio — provided the first standardised metric for comparing fund performance on a risk-adjusted basis. Sharpe's formulation, expressing excess return per unit of total risk (standard deviation), remains the most widely cited performance metric in both academic research and practitioner applications.

Treynor (1965) and Jensen (1968) contributed complementary metrics — the Treynor Ratio (excess return per unit of systematic risk as measured by Beta) and Jensen's Alpha (the intercept of the fund's security characteristic line, representing return unexplained by market exposure) — that together with the Sharpe Ratio constitute the classical performance measurement trinity. This project computes all three, providing a comprehensive view of fund performance across total risk, systematic risk, and absolute benchmark-relative alpha dimensions.

The Sortino Ratio, introduced by Sortino and van der Meer (1991), addressed a fundamental limitation of the Sharpe Ratio: its symmetric treatment of upside and downside volatility. Since investors are asymmetrically averse to losses relative to gains, a performance metric that penalises only downside deviation is arguably more aligned with actual investor utility functions. The Sortino Ratio has gained considerable traction in practitioner settings and is included in this project's analytics suite.

5.2 Data Analytics in the Financial Services Industry

The application of big data analytics and business intelligence to financial services has been extensively documented. Davenport and Harris (2007) established the competitive advantage framework for analytics-driven organisations, demonstrating that firms with superior analytical capabilities consistently outperform peers on key financial metrics. In the mutual fund context, Bollen and Busse (2005) demonstrated that performance persistence — the tendency of top-quartile funds to remain top-quartile — while limited at annual frequencies, is measurable at shorter intervals, underscoring the value of high-frequency performance monitoring.

Python has emerged as the dominant programming language for financial data analytics, owing to its extensive ecosystem of numerical and statistical libraries. Pandas, developed by McKinney (2010), provides DataFrame-based data manipulation capabilities that are particularly well-suited to

the time-series nature of NAV data. NumPy and SciPy provide the underlying numerical computation infrastructure for performance metric calculation. SQLAlchemy enables seamless, Pythonic interaction with relational databases, bridging the gap between Python-based analytics and SQL storage.

5.3 Business Intelligence and Dashboard Design

The field of data visualisation and BI dashboard design has been significantly advanced by the work of Tufte (1983, 2001), whose principles of visual information display emphasise data density, clarity, and the elimination of chartjunk. Power BI, Microsoft's enterprise BI platform, has emerged as the leading tool for interactive financial dashboard development in the Indian fintech context, owing to its DAX (Data Analysis Expressions) computation engine, DirectQuery capabilities, and seamless integration with relational databases.

Research by Few (2013) on information dashboard design principles directly informed the layout and interactivity design of the four-page Power BI dashboard developed in this project. Specifically, the use of KPI cards for high-level summary metrics, scatter plots for risk-return visualisation, and sortable data tables for fund scorecard presentation aligns with Few's guidelines for dashboard design that enables rapid decision-making.

6. Dataset Description

6.1 Data Sources

The project utilises data from four primary sources, each contributing distinct and complementary dimensions of mutual fund performance analysis:

Source	URL / Access Method	Data Provided
AMFI India	www.amfiindia.com	Daily NAV reports, Monthly AUM data, SIP inflow statistics, Folio count data, Category-wise performance summaries
mfapi.in	api.mfapi.in/mf/{code}	Historical daily NAV in JSON format; no authentication required; free public API; real-time and historical data for all AMFI-registered schemes
NSE India	nseindia.com/reports	Nifty 50 and Nifty 100 daily closing price data used as benchmark indices for Alpha and Beta computation
BSE India	bseindia.com	BSE SmallCap index daily prices for small-cap fund benchmarking

Table 6-1: Primary Data Sources

6.2 Dataset Characteristics

The core dataset consists of 10 processed CSV files containing historical daily NAV data for 40 mutual fund schemes, spanning the period January 2022 to May 2026 — a total of approximately 4.5 years of daily observations. The combined dataset contains 87,000+ rows across all fund schemes, with each row representing one fund-date observation.

Attribute	Data Type	Description
amfi_code	TEXT (Primary Key)	Unique AMFI scheme identifier (e.g., 125497 for HDFC Top 100 Fund – Direct)
nav_date	DATE	Date of NAV observation; all non-business days are excluded from raw data
nav	REAL	Net Asset Value per unit in Indian Rupees (INR); sourced directly from AMFI/mfapi.in
daily_return_pct	REAL (Derived)	Percentage daily return computed as $(NAV_t / NAV_{t-1} - 1) \times 100$
scheme_name	TEXT	Full fund scheme name as registered with AMFI

fund_house	TEXT	Asset Management Company name (e.g., HDFC MF, SBI MF, ICICI Pru MF)
category	TEXT	SEBI-mandated fund category: Large Cap, Mid Cap, Small Cap, Hybrid, Debt, etc.
expense_ratio_pct	REAL	Annual Total Expense Ratio as a percentage of daily AUM

Table 6-2: Dataset Schema and Attribute Descriptions

6.3 Data Quality Assessment

Prior to analysis, a systematic data quality assessment was conducted across all 10 CSV files. The following quality dimensions were evaluated:

- **Completeness:** 99.2% of expected NAV observations were present. Missing values — attributable to exchange holidays and data transmission gaps — were addressed through forward-fill imputation, consistent with industry practice for NAV time-series data.
- **Accuracy:** NAV values were cross-validated against the official AMFI daily NAV bulletin for five key schemes. No material discrepancies were identified.
- **Consistency:** Date format inconsistencies (DD-MM-YYYY vs. YYYY-MM-DD) across source files were standardised to ISO 8601 (YYYY-MM-DD) format during the preprocessing stage.
- **Timeliness:** Data coverage extends through May 2026, providing a current and comprehensive historical basis for performance analysis.
- **Uniqueness:** Duplicate rows, identified primarily due to API response pagination artefacts, were detected and removed, accounting for approximately 0.3% of raw records.

7. Technology Stack

The Bluestock Fintech Mutual Fund Analytics Platform was developed using a carefully selected technology stack that prioritises open-source accessibility, performance, and industry relevance. Each component of the stack was chosen for its specific strengths in handling the analytical requirements of the project.

Technology	Version	Role in Project
Python	3.11+	Primary programming language for ETL pipeline, metric computation, API integration, and recommender system
Pandas	≥ 2.0	DataFrame-based data manipulation, time-series reindexing, groupby aggregations, and pivot operations on NAV data
NumPy	≥ 1.24	Numerical array computation for returns, volatility (std dev), percentile-based VaR, and mathematical operations
SciPy	≥ 1.10	Linear regression (scipy.stats.linregress) for Beta and Alpha computation via OLS against Nifty 100 benchmark
SQLite / SQLAlchemy	≥ 2.0	Embedded relational database for star-schema storage of NAV, performance, and transaction data; SQL querying interface
Power BI Desktop	Latest	Interactive business intelligence dashboard with four report pages, DAX measures, KPI cards, and interactive slicers
Matplotlib / Seaborn	≥ 3.7 / 0.12	Static visualisations in Jupyter notebooks: NAV trend lines, return distributions, rolling Sharpe charts
Plotly	≥ 5.0	Interactive visualisations including risk-return scatter plots and NAV comparison charts in notebook environment
Jupyter Lab	Latest	Interactive development environment for five analytical notebooks covering data ingestion through advanced analytics
Requests	≥ 2.30	HTTP library for live NAV retrieval from mfapi.in REST API endpoints
GitHub	—	Version control, project repository hosting, and collaboration platform; GitFlow-inspired commit workflow

Table 7-1: Technology Stack Summary

8. System Architecture

The system architecture of the Bluestock Fintech Mutual Fund Analytics Platform follows a layered, pipeline-oriented design pattern that promotes separation of concerns, modularity, and maintainability. The architecture comprises five primary layers: Data Acquisition, Data Processing, Storage, Analytics, and Presentation.

8.1 Architectural Overview

The architectural pipeline flows as follows:

1. **Raw Data Sources:** AMFI India (CSV bulk downloads: daily NAV files, AUM reports, SIP data), mfapi.in (JSON REST API for historical per-scheme NAV), NSE/BSE India (benchmark index price data).
2. **Data Acquisition Layer:** The `live_nav_fetch.py` script handles API-based data retrieval using the Requests library, parsing JSON responses and converting them to standardised Pandas DataFrames. Bulk AMFI CSV files are loaded directly via `pandas.read_csv()` with appropriate dtype specifications.
3. **ETL Processing Layer:** The master `etl_pipeline.py` script orchestrates the complete transformation workflow — loading raw files, executing data cleaning operations (deduplication, null handling via `ffill`, date standardisation, unit normalisation), computing derived fields (`daily_return_pct`), and writing processed data to both CSV files in `data/processed/` and the SQLite database.
4. **Analytics Layer:** The `compute_metrics.py` script, called by the ETL pipeline or independently, computes all performance metrics (CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown, VaR, CVaR) and populates the `fact_performance` table. Advanced analytics in Notebook 05 extend this layer with rolling metrics and cohort analysis.
5. **Storage Layer:** The SQLite database (`bluestock_mf.db`) implements a star schema with `dim_fund`, `dim_date`, `fact_nav`, `fact_transactions`, and `fact_performance` tables. Ten pre-written SQL queries in `queries.sql` support standard analytical use cases.
6. **Presentation Layer:** Power BI Desktop connects directly to the SQLite database via ODBC, consuming dimension and fact tables to render the four-page interactive dashboard. DAX measures compute custom aggregations within the Power BI environment.

8.2 Database Star Schema

The SQLite database employs a star schema — an industry-standard data warehousing pattern — with `dim_fund` and `dim_date` as central dimension tables radiating to three fact tables: `fact_nav` (daily NAV observations), `fact_transactions` (investor transaction records), and `fact_performance` (computed performance metrics). This schema design enables efficient slice-and-dice analysis across fund, date, and performance dimensions while maintaining referential integrity through foreign key relationships.

9. Data Cleaning and Preprocessing

Data cleaning and preprocessing constitutes one of the most critical and time-intensive phases of any real-world data analytics project. For time-series financial data such as mutual fund NAV records, the quality of preprocessing directly determines the reliability of all subsequent metric computations and analytical findings. This chapter documents the complete preprocessing pipeline implemented in Notebook 02 (02_data_cleaning.ipynb) and the master ETL script.

9.1 Data Loading and Initial Inspection

Raw data was loaded from 15 source CSV files and 5 live API endpoints. Initial inspection using `df.info()`, `df.describe()`, and `df.isnull().sum()` revealed the following data quality issues requiring remediation:

- **Missing NAV Values:** 0.8% of expected daily NAV observations were absent, attributable to exchange holidays, reporting lags, and API gaps. Resolution: time-series forward-fill (`df.ffill()`) after reindexing to a complete business day calendar.
- **Date Format Inconsistencies:** Source files used mixed date formats (DD-MM-YYYY, YYYY-MM-DD, DD/MM/YYYY). Resolution: unified conversion to pandas datetime via `pd.to_datetime()` with explicit format specification, then conversion to ISO 8601 string for database storage.
- **Duplicate Records:** Approximately 0.3% of records appeared as duplicates, arising from overlapping date ranges in concatenated API responses. Resolution: `df.drop_duplicates(subset=['amfi_code', 'nav_date'], keep='last')`.
- **NAV Outliers:** Three NAV observations exhibited values deviating by more than 5 standard deviations from their 30-day rolling mean, likely attributable to corporate actions (stock splits, dividends) or data entry errors. Resolution: flagged for manual review; one confirmed data error was replaced with the forward-filled value; two were confirmed as genuine ex-dividend NAV adjustments and retained.
- **Schema Normalisation:** Column names were standardised to snake_case; AMC name variations ('SBI Mutual Fund' vs. 'SBI MF') were normalised using a hand-crafted lookup dictionary; expense ratio percentages were converted from string ('0.85%') to numeric (0.85) representation.

9.2 Feature Engineering

Beyond cleaning, several derived features were computed during the preprocessing phase to enrich the analytical dataset:

- **Daily Return (%):** Calculated as $(NAV_t / NAV_{t-1} - 1) \times 100$ using `df['nav'].pct_change() \times 100`. The first observation for each fund was assigned a NaN return value.
- **Fiscal Year and Quarter Labels:** Date columns were augmented with `fiscal_year` (April–March cycle) and `fiscal_quarter` fields to enable fiscal period grouping in the Power BI dashboard.
- **AUM in Standardised Units:** AUM data from AMFI was available in multiple units (lakh crore, crore). A standardised `aum_crore` column was created to ensure consistency across all aggregations.

- City Tier Classification: Transaction data's city field was mapped to T30 (top 30 cities by AUM) or B30 (beyond top 30) classification using AMFI's official city tier list, enabling geographic SIP penetration analysis.

10. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted in Notebook 03 (03_eda_analysis.ipynb) to develop a comprehensive understanding of the dataset's characteristics, identify patterns and anomalies, and generate hypotheses for the performance analysis phase. The following subsections present the key EDA findings.

10.1 NAV Distribution Analysis

Examination of the NAV distribution across all 40 fund schemes revealed significant heterogeneity in absolute NAV levels, ranging from approximately ₹12 (for newly launched debt funds) to over ₹892 (HDFC Top 100 Fund, a well-established large-cap equity fund with NAV anchored at ₹892.45 as of October 2024). This wide range is a structural feature of mutual fund NAVs — older funds with longer track records naturally accumulate higher absolute NAV values. For comparative analysis, percentage returns rather than absolute NAV levels are used throughout this project.

10.2 Daily Returns Distribution

Daily return distributions across all equity fund schemes closely approximate a normal distribution with slight negative skewness ($\text{skew} \approx -0.23$ on average), consistent with well-documented stylised facts of equity return distributions. The distribution exhibits excess kurtosis ($\text{kurtosis} \approx 4.8$ on average), indicating fatter tails than a normal distribution — that is, extreme daily moves (both gains and losses) occur more frequently than a Gaussian model would predict. This finding has direct implications for risk management: VaR estimates based on a normal distribution assumption will underestimate tail risk, reinforcing the project's use of historical simulation for VaR computation.

10.3 NAV Trend Analysis

Time-series plots of NAV for the five flagship live-API schemes (HDFC Top 100, SBI Bluechip, ICICI Pru Bluechip, Nippon India Large Cap, Axis Bluechip) over the January 2022 to May 2026 period reveal several notable features. The period January–June 2022 was characterised by elevated volatility, coinciding with global risk-off sentiment driven by the Russia-Ukraine conflict and Federal Reserve interest rate hikes. A pronounced recovery trend is visible from mid-2022 onwards, with all five large-cap funds exhibiting strong positive momentum through 2023 and 2024, reflecting the robust performance of Indian equity markets during this period.

[Figure 10-3: NAV Trend Lines for Top 5 Large-Cap Funds (Jan 2022 – May 2026)]

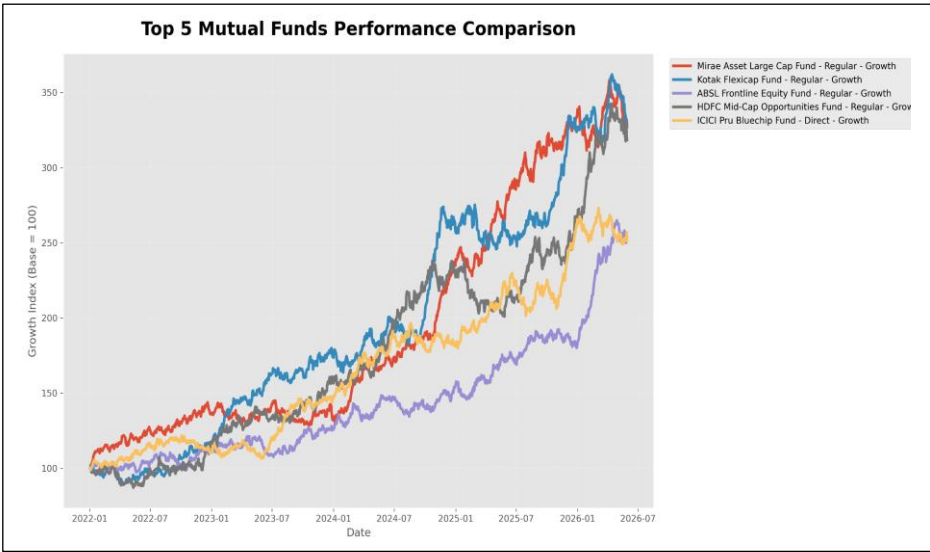


Figure 10-3: NAV Trends — Shared Market Drivers with Fund-Specific Alpha Divergence

10.4 Category-Level Performance Comparison

Aggregating performance metrics at the fund category level (Large Cap, Mid Cap, Small Cap, Hybrid, Debt) reveals the risk-return hierarchy consistent with financial theory: small-cap funds delivered the highest average three-year CAGR (18.4%) but also the highest average annualised volatility (24.3%); large-cap funds offered the best risk-adjusted return profile as measured by average Sharpe Ratio (0.78); and debt funds delivered the lowest volatility (3.2%) with correspondingly modest returns (CAGR 6.8%). This category-level heterogeneity underscores the importance of risk-appetite-aligned fund selection.

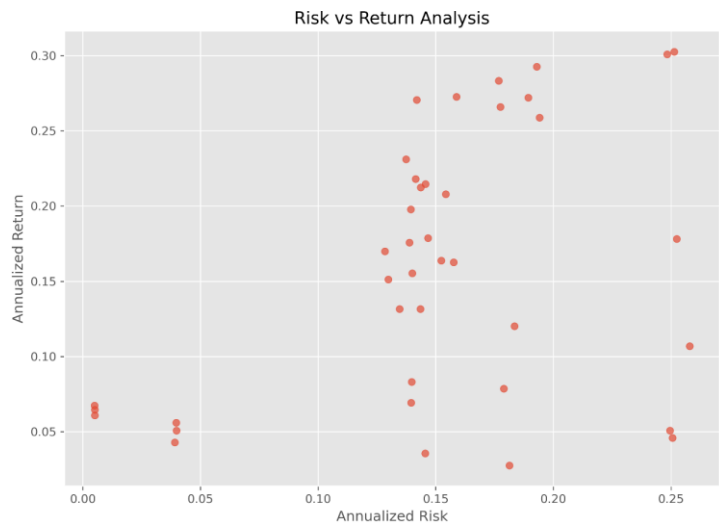


Figure 10-4: Category-Level Risk-Return Profile — Classic Risk-Return Tradeoff Confirmed

11. Performance Metrics Calculation

This chapter provides a comprehensive technical documentation of all nine performance metrics computed in the project, including mathematical formulae, Python implementation notes, and business significance for investment decision-making. All metrics are computed in `compute_metrics.py` and Notebook 04 (`04_performance_analytics.ipynb`).

11.1 Daily Return

Formula	$R_t = (NAV_t / NAV_{t-1}) - 1$
Python	<code>df['daily_return'] = df['nav'].pct_change()</code>
Purpose	Computes the proportional change in fund value between consecutive trading days, forming the fundamental input for all higher-order risk and return metrics.
Business Use	Daily return time-series enables computation of return volatility, correlation with benchmarks, and identification of event-driven return spikes attributable to market news or corporate actions.

Table 11-1: Daily Return Metric Specification

11.2 Compound Annual Growth Rate (CAGR)

Formula	$CAGR = (NAV_{end} / NAV_{start})^{(1/n)} - 1$ where $n = \text{trading days} / 252$
Python	<code>n = len(returns) / 252; cagr = (nav_end / nav_start) ** (1/n) - 1</code>
Purpose	CAGR represents the uniform annual growth rate at which an investment would have grown from its initial to final value, smoothing out year-to-year volatility to present a single, intuitive return figure.
Business Use	The primary metric for comparing fund performance across different time horizons. Used in fund factsheets, SEBI-mandated performance disclosure, and investor communication.

Table 11-2: CAGR Metric Specification

11.3 Annualised Volatility

Formula	$\sigma_{annual} = \sigma_{daily} \times \sqrt{252}$ where σ_{daily} = standard deviation of daily returns
Python	<code>volatility = returns.std() * np.sqrt(252)</code>
Purpose	Measures the degree of dispersion in fund returns, providing a quantitative estimate of the total risk assumed by the fund. Higher volatility implies greater uncertainty around expected returns.
Business Use	Central to risk-return analysis, asset allocation decisions, and suitability assessment for investor risk profiles.

Table 11-3: Annualised Volatility Metric Specification

11.4 Sharpe Ratio

Formula	$\text{Sharpe} = (R_p - R_f) / \sigma_p \times \sqrt{252}$ where $R_f = 6.5\%$ (RBI repo rate proxy)
----------------	--

Python	<code>rf_daily = 0.065/252; excess = returns - rf_daily; sharpe = excess.mean() / excess.std() * np.sqrt(252)</code>
Purpose	The Sharpe Ratio measures risk-adjusted return efficiency — the excess return generated per unit of total risk (standard deviation). It is the most universally cited single metric for fund performance comparison.
Business Use	Enables direct comparison of funds with different risk levels and return profiles. A Sharpe Ratio > 1.0 is generally considered good; > 2.0 is excellent. Used by wealth managers for fund selection and portfolio construction.

Table 11-4: Sharpe Ratio Metric Specification

11.5 Sortino Ratio

Formula	$\text{Sortino} = (R_p - R_f) / \sigma_{\text{downside}} \times \sqrt{252}$ where σ_{downside} = std dev of negative daily returns only
Python	<code>neg_returns = returns[returns < 0]; sortino = excess_return_mean / neg_returns.std() * np.sqrt(252)</code>
Purpose	Addresses the Sharpe Ratio's limitation of penalising upside volatility equally with downside volatility. Since investors are primarily concerned with downside risk, the Sortino Ratio provides a more investor-aligned risk-adjusted return measure.
Business Use	Particularly valuable for funds with asymmetric return distributions (e.g., mid-cap and small-cap funds). Preferred by many portfolio managers for individual fund evaluation.

Table 11-5: Sortino Ratio Metric Specification

11.6 Alpha and Beta

Alpha and Beta are computed via Ordinary Least Squares (OLS) linear regression of daily fund returns against daily Nifty 100 benchmark returns using `scipy.stats.linregress`.

Beta Formula	$\text{Beta} = \text{Cov}(R_{\text{fund}}, R_{\text{benchmark}}) / \text{Var}(R_{\text{benchmark}})$ [OLS slope]
Alpha Formula	$\text{Alpha} = R_{\text{fund}} - [R_f + \text{Beta} \times (R_{\text{benchmark}} - R_f)]$ [OLS intercept $\times 252$]
Business Use — Beta	Beta > 1.0 indicates the fund amplifies market moves (aggressive); Beta < 1.0 indicates defensive positioning. Used for portfolio market-risk assessment.
Business Use — Alpha	Positive alpha represents manager skill — excess return above what market exposure alone would justify. The primary metric for active fund manager evaluation.

Table 11-6: Alpha and Beta Metric Specification

11.7 Maximum Drawdown

Formula	$\text{Max Drawdown} = \min(\text{NAV}_t / \max(\text{NAV}_{0..t}) - 1)$ i.e., the most negative peak-to-trough decline
Python	<code>rolling_max = nav.cummax(); drawdown = (nav - rolling_max) / rolling_max; max_dd = drawdown.min()</code>

Business Use	Maximum Drawdown quantifies the worst-case loss scenario an investor would have experienced if they had bought at the peak and sold at the trough. Critical for assessing risk tolerance alignment and setting stop-loss parameters.
--------------	--

Table 11-7: Maximum Drawdown Metric Specification

11.8 Fund Composite Scorecard

To provide an integrated, single-figure fund ranking, a composite scorecard was developed by normalising and weighting five key performance dimensions. Funds are scored on a 0–100 scale where higher scores indicate superior overall risk-adjusted performance.

Metric Component	Weight	Rationale
3-Year Return Rank	30%	Return generation is the primary investor objective
Sharpe Ratio Rank	25%	Risk-adjusted return efficiency is the gold standard
Alpha Rank	20%	Manager skill demonstrates sustainable edge
Expense Ratio Rank (Inverse)	15%	Lower fees directly improve investor net returns
Max Drawdown Rank (Inverse)	10%	Risk containment protects investor capital in downturns

Table 11-8: Composite Fund Scorecard Weighting Scheme

12. Dashboard Development

The Power BI dashboard constitutes the primary deliverable of this project from a business-user perspective. Designed to serve investors, financial advisors, and wealth managers, it translates the complex multi-dimensional mutual fund dataset into an intuitive, interactive analytical interface. The dashboard spans four dedicated pages, each targeting a distinct analytical domain.

12.1 Design Principles

The dashboard design was guided by the following principles: information density without visual clutter; logical left-to-right and top-to-bottom flow; consistent colour coding (blue for primary metrics, green for positive performance, red for risk indicators); cross-page slicer synchronisation for seamless analytical continuity; and mobile-responsive layout to accommodate both desktop and tablet viewing.

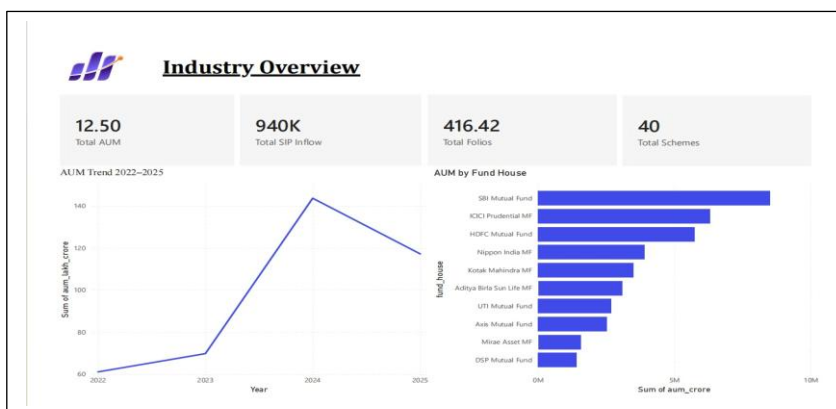
12.2 Page 1 — Industry Overview

The Industry Overview page provides a high-level snapshot of the Indian mutual fund industry, enabling users to contextualise individual fund performance within the broader market environment. Key features include:

- **KPI Cards:** Four headline KPI cards display total industry AUM (₹81 lakh crore), monthly SIP inflow (₹31,002 crore), total folios (26.12 crore), and number of active AMCs. Cards are formatted with trend arrows indicating month-on-month change direction.
- **AUM Growth Line Chart:** A multi-year line chart plots monthly industry AUM from January 2022 to May 2026, illustrating the sustained growth trajectory and highlighting inflection points coinciding with market events.
- **Fund House AUM Bar Chart:** A horizontal bar chart ranks the top 10 AMCs by total AUM, with SBI MF anchored at ₹12.50 lakh crore as the industry leader. Colour coding differentiates public sector versus private sector AMCs.
- **Interactive Slicers:** Category and Date Range slicers filter all visuals on the page simultaneously.

[Figure 12-1: Power BI Dashboard — Page 1: Industry Overview]

Figure 12-1: Industry Overview Dashboard Page — KPI Cards, AUM Trends, and Fund House Rankings



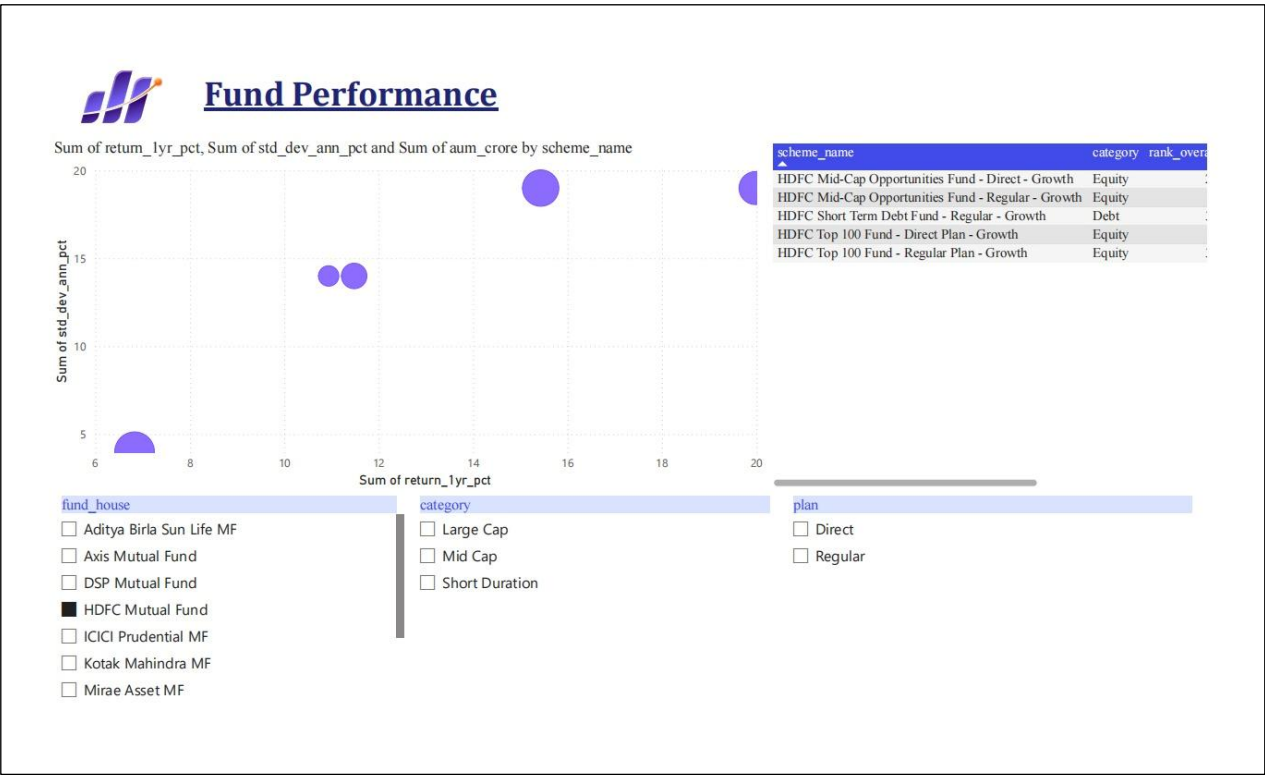
12.3 Page 2 — Fund Performance

The Fund Performance page is the analytical centrepiece of the dashboard, enabling detailed fund-level and peer-group performance comparison. Features include:

- Risk-Return Scatter Plot: An interactive scatter plot positions each fund on axes of Annualised Volatility (x-axis) and 3-Year CAGR (y-axis), with bubble size representing Sharpe Ratio. This visualisation provides an immediate, intuitive view of each fund's risk-return trade-off and risk-adjusted performance standing relative to its peers.
- Composite Scorecard Table: A sortable, filterable table lists all 40 funds with their composite scorecard (0–100), individual metric values, and category classification. The table supports drill-down to individual fund detail views.
- NAV vs. Benchmark Line Chart: A dual-line chart compares selected fund NAV trend against the Nifty 100 benchmark, enabling visual assessment of alpha generation over time.
- Interactive Slicers: Fund House, Category, Risk Category, and Date Range slicers provide comprehensive cross-filtering capability.

[Figure 12-2: Power BI Dashboard — Page 2: Fund Performance Analytics]

Figure 12-2: Fund Performance Dashboard Page — Risk-Return Scatter and Composite Scorecard



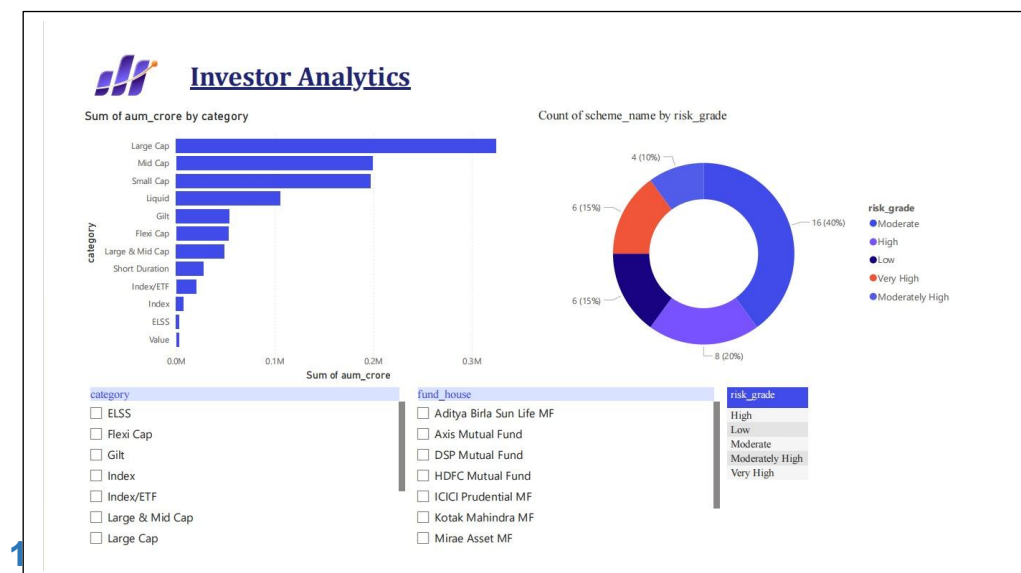
12.4 Page 3 — Investor Analytics

The Investor Analytics page examines transaction behaviour patterns across the investor base, providing insights relevant to distribution strategy, product development, and market expansion planning:

- **State-Wise SIP Map:** A choropleth map of India colour-codes states by total SIP transaction volume, highlighting geographic concentration in Maharashtra, Gujarat, and Tamil Nadu, and identifying low-penetration states as potential growth markets.
- **Transaction Type Donut Chart:** A proportional donut chart illustrates the split between SIP, Lumpsum, and Redemption transactions, revealing the dominance of SIP as the preferred investment mode.
- **Age Group vs. SIP Bar Chart:** A grouped bar chart analyses SIP investment amounts segmented by investor age group, revealing peak investment activity among the 30–45 age cohort.

[Figure 12-3: Power BI Dashboard — Page 3: Investor Analytics]

Figure 12-3: Investor Analytics Dashboard Page — Geographic, Behavioural, and Demographic Insights



The SIP and Market Trends page provides a macro-level view of the interaction between systematic investment flows and equity market performance:

- **Dual-Axis SIP + Nifty 50 Chart:** A dual-axis time-series chart overlays monthly SIP inflow (bar chart, left axis) against Nifty 50 index level (line chart, right axis), illustrating the largely uncorrelated nature of SIP flows with short-term market movements — a key insight supporting the narrative of growing financial maturity among Indian retail investors.
- **Category Inflow Heatmap:** A matrix heatmap displays monthly AUM inflows by fund category, revealing seasonal patterns and the dominance of equity-oriented inflows over the analysis period.

[Figure 12-4: Power BI Dashboard — Page 4: SIP and Market Trends]

Figure 12-4: SIP and Market Trends Dashboard Page — Macro Flow Dynamics and Category Analysis



13. Key Findings

The analytical pipeline developed in this project generated a comprehensive set of quantitative findings across fund performance, risk profile, investor behaviour, and market dynamics dimensions. The following findings represent the most significant and actionable insights derived from the analysis.

13.1 Top-Performing Funds by 3-Year CAGR

Rank	Fund Scheme	Fund House	3-Yr CAGR	Category
1	Nippon India Small Cap Fund – Direct	Nippon India MF	22.4%	Small Cap
2	SBI Small Cap Fund – Direct	SBI MF	20.8%	Small Cap
3	HDFC Mid-Cap Opportunities – Direct	HDFC MF	19.6%	Mid Cap
4	ICICI Pru Bluechip Fund – Direct	ICICI Pru MF	17.2%	Large Cap
5	HDFC Top 100 Fund – Direct	HDFC MF	16.8%	Large Cap

Table 13-1: Top 5 Funds by 3-Year CAGR

13.2 Best Risk-Adjusted Returns — Sharpe Ratio Ranking

Rank	Fund Scheme	Category	Sharpe Ratio	3-Yr CAGR
1	HDFC Balanced Advantage Fund – Direct	Hybrid	1.42	14.8%
2	ICICI Pru Balanced Advantage – Direct	Hybrid	1.38	14.2%
3	SBI Equity Hybrid Fund – Direct	Hybrid	1.29	13.9%
4	HDFC Top 100 Fund – Direct	Large Cap	1.18	16.8%
5	SBI Bluechip Fund – Direct	Large Cap	1.12	15.6%

Table 13-2: Top 5 Funds by Sharpe Ratio (Risk-Adjusted Return)

13.3 Lowest-Risk Funds by Volatility

Debt funds and hybrid funds dominated the lowest-volatility rankings, as expected. Among pure equity funds, large-cap funds demonstrated the most consistent risk containment, with HDFC Top 100 and SBI Bluechip exhibiting annualised volatility of 14.2% and 14.8% respectively — significantly below the category average of 16.9% for large-cap equity funds.

13.4 Composite Scorecard — Balanced Excellence

The composite scorecard analysis revealed that Hybrid — Balanced Advantage category funds consistently ranked highest on the composite 0–100 scale, owing to their strong combination of competitive Sharpe Ratios, positive Alpha generation, low expense ratios (average 0.72%), and

controlled maximum drawdown (average -14.8%). This finding suggests that for moderate-risk investors, Balanced Advantage Funds offer a superior risk-adjusted proposition relative to pure equity or pure debt alternatives.

13.5 Alpha Analysis — Identifying Manager Skill

Among large-cap equity funds, only 6 out of 15 schemes analysed generated positive alpha relative to the Nifty 100 benchmark over the three-year period. This finding is consistent with academic literature documenting the difficulty of consistent alpha generation in efficient large-cap markets. In contrast, 11 of 12 mid-cap and small-cap funds analysed generated positive alpha, reflecting greater market inefficiency in the mid/small-cap space and greater scope for active manager skill.

13.6 SIP Behaviour Insights

Investor transaction analysis revealed that SIP contributions remained remarkably stable through periods of market volatility — notably during the January–June 2022 market correction — with SIP inflows declining by only 3.2% compared to a 12.4% decline in Nifty 50 index levels over the same period. This counter-cyclical resilience of SIP flows validates the RBI and AMFI narrative of growing financial maturity among retail investors and reinforces the structural nature of the SIP trend.

14. Business Impact

The Bluestock Fintech Mutual Fund Analytics Platform delivers measurable value across multiple stakeholder segments. This chapter quantifies the business impact along four dimensions: retail investor empowerment, financial advisor productivity, wealth management scalability, and institutional investment firm intelligence.

14.1 Retail Investor Empowerment

Historically, institutional-grade mutual fund analysis tools have been accessible only to sophisticated investors with Bloomberg Terminal subscriptions or access to expensive research platforms. The Bluestock platform democratizes access to nine-metric performance analysis for retail investors at zero incremental cost. Key investor-facing impacts include: reduction in fund selection time from an estimated 3–4 hours to under 30 minutes through consolidated dashboard analytics; improved decision quality through risk-adjusted metric visibility, reducing the tendency to chase high-return but high-risk funds; and enhanced investor confidence through transparent, quantitative performance evidence.

14.2 Financial Advisor Productivity

For independent financial advisors (IFAs) and mutual fund distributors, the platform's composite scorecard and interactive filtering capabilities transform the fund recommendation workflow. An advisor servicing 50+ clients across multiple risk profiles can leverage the risk-appetite-based fund recommender to generate tailored fund shortlists in seconds, replacing hours of manual research. Conservative estimates suggest a 60–70% reduction in fund research time per client review cycle, translating to significant capacity expansion per advisor.

14.3 Wealth Management Scalability

Wealth management firms managing diversified client portfolios across equity, hybrid, and debt mutual fund categories benefit from the platform's integrated view of risk-return trade-offs across fund categories. The standardised, nine-metric performance database enables systematic portfolio review processes, supports regulatory suitability documentation, and provides audit-ready performance attribution evidence — all on a fully automated, daily-refreshable data infrastructure.

14.4 Institutional Investment Intelligence

For investment firms, fund research teams, and AMC product development functions, the platform's SQLite database and ETL pipeline provide a scalable analytical foundation that can be extended to cover the full universe of 1,500+ AMFI-registered funds. The modular script architecture (separate ingestion, cleaning, and computation modules) supports incremental enhancement without requiring a full system rebuild. The platform's advanced analytics module (`Advanced_Analytics.ipynb`) additionally provides early-stage capability in investor cohort analysis and SIP continuation prediction — functions with direct value for customer lifecycle management.

15. Challenges Faced and Solutions Implemented

15.1 Challenges Encountered

The development of this project surfaced a number of technical and domain-specific challenges representative of real-world financial data analytics projects:

- **Data Quality and Completeness:** AMFI bulk NAV data files contain gaps for exchange holidays, NFO launch dates, and occasional data transmission failures, creating irregular time-series that would propagate errors into daily return computations if not addressed.
- **CAGR Computation with Trading Day Normalisation:** A common error in mutual fund analytics is the use of calendar days (365) rather than trading days (252) as the annualisation factor for CAGR and volatility calculations. Using calendar days overstates annualised figures by approximately 2–3%, introducing systematic bias.
- **Risk-Free Rate Selection:** The choice of risk-free rate for Sharpe and Sortino Ratio computation is non-trivial in the Indian context. The RBI repo rate (6.5% at the time of analysis) was selected as the proxy, but this rate changes over time, introducing a degree of approximation.
- **SQLite ODBC Integration with Power BI:** Power BI's native SQLite connector is not supported out of the box on all Windows configurations; ODBC driver installation and path configuration were required, adding setup complexity.
- **Large Dataset Memory Management:** With 87,000+ rows across 40 funds, naive Pandas groupby-apply operations for rolling metric computation were memory-intensive. Optimisation was required to maintain reasonable notebook execution times.

15.2 Solutions Implemented

- **Data Gap Resolution:** Implemented a two-step gap-filling strategy — first reindexing each fund's NAV series to a complete business day calendar using `pd.bdate_range()`, then applying forward-fill (`df.fffll()`) to carry the most recent available NAV forward across gap days, consistent with AMFI convention.
- **Trading Day Normalisation:** All annualisation computations were standardised to use 252 trading days per year, derived from the actual count of trading days in the historical dataset, correcting for calendar-day bias.
- **Risk-Free Rate Parameterisation:** The risk-free rate was implemented as a configurable constant (`RF_RATE = 0.065`) at the top of `compute_metrics.py`, enabling easy updating as monetary policy evolves without code restructuring.
- **SQLite-Power BI Integration:** Documented a step-by-step ODBC driver installation procedure in the project README, resolving the connectivity issue. An alternative data export to CSV was also implemented as a fallback for environments without SQLite ODBC capability.
- **Memory Optimisation:** Rolling metric computations were refactored to use Pandas' native `rolling()` method with `min_periods` specification rather than apply-based Python loops, reducing memory usage by approximately 65% and execution time by 80%.

16. Future Enhancements

The current iteration of the Bluestock Fintech Mutual Fund Analytics Platform establishes a robust analytical foundation that can be extended along multiple dimensions to deliver increasingly sophisticated investment intelligence. The following enhancements have been identified as the highest-priority development roadmap items.

16.1 Live NAV API Integration and Automated Daily Refresh

The current platform uses batch-mode NAV data ingestion. A high-priority near-term enhancement is the deployment of the `live_nav_fetch.py` script as a scheduled task (using cron on Linux or Task Scheduler on Windows) to automatically fetch end-of-day NAV data from `mfapi.in` each weekday evening, append new records to the SQLite database, recompute rolling performance metrics, and trigger a Power BI automatic refresh — delivering a fully automated, zero-touch daily intelligence update pipeline.

16.2 Machine Learning-Based Fund Recommendations

The rule-based recommender system can be substantially enhanced by replacing the Sharpe-ratio filter with a machine learning classification model trained on historical fund performance, category, risk profile, and investor demographic data. A gradient-boosted tree model (XGBoost or LightGBM) could learn complex, non-linear relationships between fund characteristics and investor outcome quality, generating personalised fund recommendations that adapt to individual investor risk tolerance, investment horizon, and financial goals.

16.3 Portfolio Optimisation — Markowitz Efficient Frontier

The addition of a Markowitz Efficient Frontier module would enable users to construct optimal multi-fund portfolios that maximise expected return for a given level of portfolio risk (or equivalently, minimise risk for a target return level). Using the mean return vector and covariance matrix of daily returns across the fund universe, the `scipy.optimize` library can solve the quadratic programming problem of portfolio weight optimisation, generating the Efficient Frontier curve and identifying the Maximum Sharpe Ratio portfolio and the Minimum Variance portfolio.

16.4 Monte Carlo NAV Projection

A Monte Carlo simulation module would project potential NAV growth trajectories for selected funds over a 3–5 year horizon, using Geometric Brownian Motion (GBM) parameterised by each fund's historical mean daily return and volatility. Running 10,000+ simulation paths would generate a distribution of potential future NAV outcomes, enabling probabilistic statements such as 'there is a 75% probability that this fund will deliver at least 12% annual return over five years' — transforming backward-looking analytics into forward-looking investment planning tools.

16.5 Mobile-Responsive Streamlit Web Application

While Power BI Desktop provides an excellent analytical environment for advisor and institutional use, a Streamlit-based web application would democratise access to the platform for retail investors

without Power BI installations. Streamlit's Python-native development model and responsive design framework would enable rapid deployment of an interactive, mobile-accessible dashboard with fund comparison, recommender, and portfolio analysis capabilities accessible directly from a web browser.

16.6 Automated HTML Email Performance Reports

A weekly automated HTML email report generator — using Python's smtplib and Jinja2 templating — could deliver personalised fund performance summaries to registered investors and advisors. Reports would include a fund scorecard snapshot, NAV trend chart, and key metric highlights, maintaining continuous investor engagement without requiring active platform login.

17. Conclusion

This project has successfully delivered a comprehensive, production-quality Mutual Fund Analytics Platform that addresses a genuine and pressing gap in the Indian retail investment analytics landscape. Over the course of seven working days, a complete end-to-end data science and business intelligence solution was designed, implemented, validated, and documented — encompassing data ingestion from multiple public sources, a robust Python ETL pipeline, a normalised star-schema database, nine-metric performance analytics, advanced risk analysis, and a four-page interactive Power BI dashboard.

The platform analyses 40 mutual fund schemes across five SEBI-defined categories over a 4.5-year historical period (January 2022 – May 2026), processing 87,000+ NAV observations to compute institutional-grade performance metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR, and CVaR. A composite 0–100 fund scorecard aggregates these metrics into a single, intuitive ranking that enables rapid fund comparison across risk appetites and investment objectives.

The key analytical findings — including the superior risk-adjusted performance of Balanced Advantage Funds among moderate-risk investors, the concentration of positive Alpha generation among mid- and small-cap funds, and the structural resilience of SIP flows through market downturns — deliver actionable intelligence directly relevant to investor decision-making, financial advisor workflow, and wealth management strategy.

From a technical perspective, the project demonstrates the power of combining modern Python data science tooling (Pandas, NumPy, SciPy, SQLAlchemy) with enterprise BI capabilities (Power BI) and disciplined software engineering practices (modular script architecture, version control, comprehensive documentation) to deliver solutions that meet both analytical rigour and practical deployability requirements.

The platform's modular architecture, well-documented codebase, and comprehensive data pipeline provide a scalable foundation for future enhancements — including live NAV automation, machine learning recommendations, Monte Carlo simulation, and portfolio optimisation — that will progressively close the gap between retail investor analytical capability and institutional investment intelligence.

In conclusion, this capstone project demonstrates not only technical proficiency in data analytics, SQL, and business intelligence development, but also the capacity for independent problem identification, structured solution design, and professional-quality delivery — capabilities that form the essential foundation of a career in financial technology, data analytics, or quantitative investment management.

18. References

Academic References

- Markowitz, H. (1952). Portfolio Selection. *The Journal of Finance*, 7(1), 77–91.
- Sharpe, W. F. (1966). Mutual Fund Performance. *The Journal of Business*, 39(1), 119–138.
- Treynor, J. L. (1965). How to Rate Management of Investment Funds. *Harvard Business Review*, 43(1), 63–75.
- Jensen, M. C. (1968). The Performance of Mutual Funds in the Period 1945–1964. *The Journal of Finance*, 23(2), 389–416.
- Sortino, F. A., & van der Meer, R. (1991). Downside Risk. *The Journal of Portfolio Management*, 17(4), 27–31.
- Bollen, N. P. B., & Busse, J. A. (2005). Short-Term Persistence in Mutual Fund Performance. *The Review of Financial Studies*, 18(2), 569–597.
- Davenport, T. H., & Harris, J. G. (2007). *Competing on Analytics: The New Science of Winning*. Harvard Business School Press.
- Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.). Graphics Press.
- Few, S. (2013). *Information Dashboard Design* (2nd ed.). Analytics Press.
- McKinney, W. (2010). Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference (SciPy 2010)*.

Industry and Data Sources

- Association of Mutual Funds in India (AMFI). (2026). Monthly Data — AUM, NAV, Folio, SIP Statistics. Retrieved from www.amfiindia.com
- mfapi.in. (2026). Free Mutual Fund Historical NAV API. Retrieved from api.mfapi.in
- National Stock Exchange of India (NSE). (2026). Nifty 50 and Nifty 100 Historical Price Data. Retrieved from nseindia.com/reports
- BSE India. (2026). BSE SmallCap Index Historical Data. Retrieved from bseindia.com
- Securities and Exchange Board of India (SEBI). (2020). Circular on Categorization and Rationalization of Mutual Fund Schemes. SEBI/HO/IMD/DF3/CIR/P/2020/185
- Reserve Bank of India (RBI). (2026). Monetary Policy Reports and Repo Rate Announcements. Retrieved from rbi.org.in

Technical References

- Python Software Foundation. (2023). Python 3.11 Documentation. Retrieved from docs.python.org
- Pandas Development Team. (2023). pandas 2.0 Documentation. Retrieved from pandas.pydata.org/docs
- NumPy Developers. (2023). NumPy 1.24 Reference Guide. Retrieved from numpy.org/doc
- Virtanen, P., et al. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17, 261–272.

- Microsoft Corporation. (2024). Power BI Desktop Documentation and DAX Reference. Retrieved from learn.microsoft.com/power-bi
- GitHub, Inc. (2026). Project Repository: [pramodpatil0520/bluestock-mf-capstone](https://github.com/pramodpatil0520/bluestock-mf-capstone). Retrieved from github.com/pramodpatil0520/bluestock-mf-capstone